

Radxa Cubie A7A Camera Monitor Setup Guide

TM Seating Sanand - installation, operation, recording, and troubleshooting handover document

This guide explains how to set up the TM Seating Sanand camera monitor on a Radxa Cubie A7A from a fresh board. It is written for a new technician who may not have prior Linux, Radxa, MediaMTX, or RTSP experience.

1. System Overview

The system has four main parts:

```
IP Camera
-> RTSP stream over Ethernet
-> MediaMTX running on Radxa
-> Browser live preview through WebRTC/HLS
-> Python camera monitor app records RTSP locally with ffmpeg
```

The browser must not open the camera RTSP URL directly. Chrome/Chromium displays the stream through MediaMTX:

```
Live preview through MediaMTX:
http://127.0.0.1:8889/pramacam

Main application page:
http://127.0.0.1:8080
```

From another laptop on the same Wi-Fi network, replace 127.0.0.1 with the Radxa Wi-Fi IP address.

2. Required Hardware

Use the following hardware:

- Radxa Cubie A7A board
- Radxa-compatible power adapter, preferably 5V/3A or better
- microSD card with Radxa Debian image installed
- HDMI monitor
- USB keyboard and mouse
- Ethernet cable
- IP camera
- Camera power supply, PoE injector, or PoE switch, depending on camera type
- Wi-Fi network for remote access or app viewing from laptop

Important: if the camera is a PoE camera, the Radxa Ethernet port cannot power it. Use a PoE switch or PoE injector.

3. Fixed IP Plan

Use this IP plan unless the site network requires a different one:

```
Camera IP:          192.168.10.20
Radxa Ethernet IP:  192.168.10.50
Radxa Wi-Fi IP:     Assigned by Wi-Fi router, for example 192.168.1.51
MediaMTX local RTSP: rtsp://127.0.0.1:8554/pramacam
MediaMTX WebRTC page: http://127.0.0.1:8889/pramacam
Main app page:      http://127.0.0.1:8080
```

The camera and Radxa Ethernet must be in the same range:

```
192.168.10.x
```

The Wi-Fi IP may be different, for example:

```
192.168.1.x
```

This is normal. Use Wi-Fi IP for accessing the app from another laptop. Use Ethernet IP for camera communication.

4. First Boot Checks

Open a terminal on the Radxa:

```
Ctrl + Alt + T
```

Check the IP addresses:

```
hostname -I
```

Check network connections:

```
nmcli con show
nmcli device status
```

If the terminal asks for a password after a sudo command, enter the Radxa password. The password will not show while typing.

5. Set Permanent Ethernet IP

First find the exact Ethernet connection name:

```
nmcli con show
```

The name is usually:

```
Wired connection 1
```

Linux is case-sensitive. `Wired Connection 1` is not the same as `Wired connection 1`.

Set the Ethernet IP permanently:

```
sudo nmcli con mod "Wired connection 1" ipv4.method manual ipv4.addresses 192.168.10.50/24
sudo nmcli con mod "Wired connection 1" ipv4.gateway ""
sudo nmcli con mod "Wired connection 1" ipv4.dns ""
sudo nmcli con down "Wired connection 1"
sudo nmcli con up "Wired connection 1"
```

Verify:

```
hostname -I
```

You should see:

```
192.168.10.50
```

Test the camera:

```
ping 192.168.10.20
```

Stop ping:

```
ctrl + c
```

If you see `Destination Host Unreachable`, check the Ethernet cable, camera power, PoE injector/switch, and camera IP address.

6. Install Required Packages

Install Git and ffmpeg:

```
sudo apt update  
sudo apt install git ffmpeg -y
```

Verify ffmpeg:

```
ffmpeg -version
```

If the app shows `No such file or directory: 'ffmpeg'`, ffmpeg is missing or not installed correctly.

7. Download Project Code

Go to the home directory:

```
cd ~
```

Clone the project from GitHub:

```
git clone https://github.com/YOUR_USERNAME/YOUR_REPO_NAME.git TM_Seating_sanand
```

Enter the project folder:

```
cd ~/TM_Seating_sanand
```

If the project already exists and you only need to update it:

```
cd ~/TM_Seating_sanand  
git pull
```

Note: if the GitHub repository is private, GitHub requires a Personal Access Token instead of the normal account password.

8. Install MediaMTX

Use MediaMTX to bridge the camera RTSP stream into browser-friendly WebRTC/HLS.

For Radxa Cubie A7A, check CPU type:

```
uname -m
```

If it shows:

```
aarch64
```

download:

```
mediamtx_v1.11.2_linux_arm64v8.tar.gz
```

Manual download page:

```
https://github.com/bluenvoron/mediamtx/releases/tag/v1.11.2
```

Place the extracted MediaMTX folder inside:

```
~/TM_Seating_sanand/
```

Expected folder:

```
~/TM_Seating_sanand/mediamtx_v1.11.2_linux_arm64v8
```

Make the executable runnable:

```
cd ~/TM_Seating_sanand/mediamtx_v1.11.2_linux_arm64v8
chmod +x mediamtx
```

9. Configure MediaMTX

Open the MediaMTX config:

```
cd ~/TM_Seating_sanand/mediamtx_v1.11.2_linux_arm64v8
nano mediamtx.yml
```

Find the paths : section and configure the path name as exactly:

```
paths:
  pramacam:
    source: rtsp://USERNAME:PASSWORD@192.168.10.20:554/STREAM_PATH
```

Replace:

```
USERNAME
PASSWORD
STREAM_PATH
```

with the real camera username, password, and stream path.

Common stream paths:

```
Hikvision main stream: /Streaming/Channels/101
Hikvision sub stream: /Streaming/Channels/102
Dahua main stream: /cam/realmonitor?channel=1&subtype=0
Dahua sub stream: /cam/realmonitor?channel=1&subtype=1
```

Important: the path must be pramacam, not paracam. The app records from:

```
rtsp://127.0.0.1:8554/pramacam
```

Save in nano:

```
Ctrl + O
Enter
Ctrl + X
```

10. Start MediaMTX

Open terminal 1:

```
cd ~/TM_Seating_sanand/mediamtx_v1.11.2_linux_arm64v8
./mediamtx
```

Keep this terminal open.

Expected successful signs:

```
MediaMTX v1.11.2
configuration loaded
[path pramacam] [RTSP source] started
```

If you see:

```
listen udp :8000: bind: address already in use
```

another MediaMTX process is already running. Stop it:

```
pkill mediamtx
./mediamtx
```

Test live stream in Chromium:

```
http://127.0.0.1:8889/pramacam
```

If you see the camera footage here, MediaMTX is working.

11. Start the Camera Monitor App

Open terminal 2:

```
cd ~/TM_Seating_sanand
python3 server.py
```

Expected output:

```
Camera Monitor running at http://0.0.0.0:8080
Recording source: rtsp://127.0.0.1:8554/pramacam
Recordings folder: /home/radxa/TM_Seating_sanand/recordings
```

Retention: 2 day(s), maximum 2 day(s)

Open the app on the Radxa:

```
http://127.0.0.1:8080
```

Open the app from another laptop on the same Wi-Fi:

```
http://RADXA_WIFI_IP:8080
```

Example:

```
http://192.168.1.51:8080
```

Do not open `index.html` directly. If the browser shows HTML code, you opened the file directly. Use:

```
http://127.0.0.1:8080
```

12. Recording Behavior

The app records from:

```
rtsp://127.0.0.1:8554/pramacam
```

Default segment length:

```
30 minutes
```

This means the app creates one MP4 file every 30 minutes:

```
recordings/data_YYYY-MM-DD_HH-MM-SS.mp4
```

A recording may not play while it is still being written. Wait until the 30-minute segment finishes before testing playback.

The app prepares browser playback using HLS cache under:

```
recordings/.hls
```

If playback fails for old or broken files, clear old recordings:

```
cd ~/TM_Seating_sanand
rm recordings/*.mp4
rm -rf recordings/.hls
```

Then restart MediaMTX and the app.

13. Recording Size and Storage Planning

Measured Radxa recording size:

```
30 minutes = about 950 MB
1 hour     = about 1.9 GB
24 hours   = about 45.6 GB
2 days     = about 91.2 GB
```

Recommendation:

```
64 GB card: keep 1 day
128 GB card: keep 2 days, but monitor free space
```

If Raspberry Pi recordings were smaller, for example 350 MB per 30 minutes, the camera stream used there was probably lower bitrate or sub stream. The app uses `ffmpeg -c copy`, so it saves the stream exactly as received.

To reduce file size while keeping 30-minute recordings, change the camera stream or bitrate:

```
Target bitrate: 1500 kbps to 1800 kbps for about 350 MB per 30 minutes
```

For Hikvision, use sub stream:

```
/Streaming/Channels/102
```

For Dahua, use sub stream:

```
/cam/realmonitor?channel=1&subtype=1
```

14. Normal Startup Order

Always start in this order:

1. Camera powered on and connected
2. Radxa Ethernet active at 192.168.10.50
3. MediaMTX running
4. Camera monitor app running
5. Browser opened at `http://127.0.0.1:8080`

Commands:

Terminal 1:

```
cd ~/TM_Seating_sanand/mediamtx_v1.11.2_linux_arm64v8
./mediamtx
```

Terminal 2:

```
cd ~/TM_Seating_sanand
python3 server.py
```

Browser:

```
http://127.0.0.1:8080
```

15. Troubleshooting

Browser shows HTML code

Cause: `index.html` was opened directly.

Fix:

```
Open http://127.0.0.1:8080
```

Live works in MediaMTX but app has no video

Check that MediaMTX path is exactly:

```
pramacam
```

Test:

```
http://127.0.0.1:8889/pramacam
```

If MediaMTX config says paracam, change it to pramacam.

ffmpeg says 404 Not Found

Error:

```
rtsp://127.0.0.1:8554/pramacam: Server returned 404 Not Found
```

Cause: MediaMTX does not have a path named pramacam, or MediaMTX is not running.

Fix:

```
pkill mediamtx
cd ~/TM_Seating_sanand/mediamtx_v1.11.2_linux_arm64v8
./mediamtx
```

Confirm it says:

```
[path pramacam]
```

ffmpeg says connection refused

Error:

```
Connection to tcp://127.0.0.1:8554 failed: Connection refused
```

Cause: MediaMTX is stopped.

Fix: start MediaMTX first, then start the Python app.

Could not start ffmpeg

Error:

```
No such file or directory: 'ffmpeg'
```

Fix:

```
sudo apt update
sudo apt install ffmpeg -y
```


MediaMTX says UDP 8000 already in use

Error:

```
listen udp :8000: bind: address already in use
```

Fix:

```
pkill mediampx  
./mediampx
```

Camera ping fails

Command:

```
ping 192.168.10.20
```

Possible causes:

- Camera is not powered
- PoE injector or PoE switch is missing
- Ethernet cable is loose
- Camera IP is not 192.168.10.20
- Radxa Ethernet IP is not 192.168.10.50

Check:

```
hostname -I  
nmcli device status
```

Recording playback unavailable

Possible causes:

- The 30-minute MP4 is still being written
- The file is old/broken from a previous failed run
- MediaMTX was stopped during recording
- ffmpeg could not prepare HLS playback

Check logs:

```
cd ~/TM_Seating_sanand  
tail -80 recordings/recorder.log  
ls -lh recordings
```

Clear broken recordings:

```
rm recordings/*.mp4  
rm -rf recordings/.hls
```

Restart MediaMTX first, then the app.

16. Useful Commands

Check Radxa IP:

```
hostname -I
```

Check network connections:

```
nmcli con show  
nmcli device status
```

Test camera:

```
ping 192.168.10.20
```

Stop ping:

```
ctrl + c
```

Stop MediaMTX:

```
pkill mediamtx
```

Stop Python server:

```
ctrl + c
```

Check recordings:

```
ls -lh ~/TM_Seating_sanand/recordings
```

Check recorder log:

```
tail -80 ~/TM_Seating_sanand/recordings/recorder.log
```

Check free disk space:

```
df -h
```

17. Final Verification Checklist

Before handing over the system, verify:

- Radxa boots normally
- Ethernet IP remains 192.168.10.50 after reboot
- Camera responds to ping 192.168.10.20
- MediaMTX starts without errors
- `http://127.0.0.1:8889/pramacam` shows live camera
- `python3 server.py` starts without ffmpeg errors
- `http://127.0.0.1:8080` opens the TM camera monitor page
- Live view works in the app

- A completed 30-minute recording appears in Recordings
- Completed recording plays in the browser
- Retention is set according to available storage

End of setup guide